A Counting Bound on Transverse Relaxation: Schmidt-Rank Growth, the Margolus–Levitin Ceiling, and a Perturbation-Independent Loschmidt-Echo Floor

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Abstract

The transverse relaxation time T_2 of a spin ensemble, and the irreducible attenuation of its Hahn and Loschmidt echoes, are normally computed from a microscopic relaxation theory (Bloch–Redfield, or its strong-coupling Anderson–Kubo limit) in which the relaxation rate is a second-moment integral of a bath correlation function. We present an independent, information-theoretic derivation that does not begin from a correlation function but from a single counting principle: the number of mutually distinguishable states a system can resolve under a Hamiltonian of energy spread ΔE grows no faster than the Margolus–Levitin ceiling $2\Delta E/\pi\hbar$. We make this concrete by identifying the "count" with the effective Schmidt rank $M_{\text{eff}}(t) = \exp S_{\text{vN}}(\rho_{\text{S}}(t))$ of the reduced spin state, where S_{vN} is the von Neumann entropy of the spin density operator after the bath is traced out. Three results follow. **(i)** A *counting bound on relaxation*: the transverse relaxation rate is bounded by the rate of Schmidt-rank growth, which is in turn bounded by the Margolus–Levitin ceiling evaluated at the spin–bath coupling energy; saturation of the bound reproduces the Anderson–Kubo rigid-lattice rate $1/T_2 \simeq \langle\delta\omega^2\rangle^{1/2}$ with no fitted constant. **(ii)** A *derivation, not a postulate, of monotonicity*: starting from a strictly time-reversal-invariant total Hamiltonian, the reduced spin entropy $S_{\text{vN}}(\rho_{\text{S}}(t))$ is non-decreasing under coarse-grained evolution by the data-processing inequality, so the count is monotone although the microscopic dynamics is reversible—the arrow enters solely through the partial trace, and the spin echo separates the reversible (product-state) part of dephasing from the irreversible (entanglement-committed) part exactly. **(iii)** A *perturbation-independent echo floor*: counting the distinguishable many-body trajectories accessible after n coherent steps over d coupling channels gives $T(n, d) = d(d + 1)^{n-1}$, whose logarithmic growth rate $\ln(d+1)$ per step is independent of the coupling strength; converted to a physical rate this predicts a Loschmidt-echo decay floor $1/\tau_{\text{floor}} = (\omega_{\text{loc}}/2\pi)\ln(d+1)$ that does not recede as the reversal engineering improves. The prediction reproduces the observed environment-independence of polarization-echo decay and, unlike the semiclassical Lyapunov account, predicts a nonzero floor for non-chaotic and integrable many-body spin systems—a falsifiable point of departure. We state the assumptions explicitly, prove each result, and identify the experiments that would confirm or refute the framework.

Keywords: transverse relaxation; spin echo; Loschmidt echo; Schmidt rank; von Neumann entropy; Margolus–Levitin bound; quantum speed limit; data-processing inequality; environment-independent decoherence; irreversibility.

Contents

1 Introduction

1.1 The question

A precessing spin ensemble in a static magnetic field loses transverse coherence on two timescales (Abragam, 1961; Bloch, 1946; Slichter, 1990). The faster, T_2^*, reflects static spread in precession frequencies across the sample; it is reversible, and the Hahn echo (Hahn, 1950) undoes it exactly with a single refocusing pulse. The slower, T_2, reflects time-dependent fluctuating fields; it is not undone by refocusing, and it sets the floor of the echo amplitude. The standard account of T_2 is Bloch–Redfield theory (Abragam, 1961; Redfield, 1957): the relaxation rate is a spectral integral of a bath autocorrelation function, reducing to $1/T_2 = \frac{1}{2}\gamma^2\langle\delta B^2\rangle\tau_c$ in the motional-narrowing limit and to the rigid-lattice second moment $1/T_2 \simeq \langle\delta\omega^2\rangle^{1/2}$ when the bath is slow (Anderson and Weiss, 1953; Kubo, 1962).

This theory is quantitatively successful and we do not dispute it. We ask a different question. The relaxation rate, in the standard account, is the output of a correlation function whose form is supplied by the microscopic model. Is there a derivation of the *same number* from a principle that does not presuppose the correlation function—a principle about how fast distinguishable information can be generated at all? And does that principle say anything the correlation-function account does not, in the regime where the two might diverge?

We answer both questions affirmatively. The organizing principle is a *quantum speed limit*: the Margolus–Levitin theorem (Deffner and Campbell, 2017; Mandelstam and Tamm, 1945; Margolus and Levitin, 1998), which bounds the rate at which a system can pass between distinguishable states by its energy spread. We convert this into a bound on a *counting* quantity—the effective number of states the environment has resolved about the spins—and show that the relaxation rate is the saturation of this bound.

1.2 The three layers of the argument

The paper has a deliberate three-layer structure, each layer answering a distinct objection one might raise against the previous one.

Foundation (Section 3): a counting ceiling fixes the rate constant. We define a count—the effective Schmidt rank of the reduced spin state—and prove that its rate of growth is bounded by the Margolus–Levitin ceiling evaluated at the spin–bath interaction energy. Saturating the ceiling reproduces the Anderson–Kubo rigid-lattice relaxation rate with no adjustable proportionality constant. This answers the objection "the rate constant is fitted, not derived."

Mechanism (Section 4): monotonicity is derived, not assumed. We then show *why* the count grows at all, and only forward. Starting from a strictly time-reversal-invariant total Hamiltonian for spins-plus-bath, the reduced spin entropy is non-decreasing under the coarse-grained (bath-averaged) dynamics by the data-processing inequality for completely positive trace-preserving maps. The microscopic dynamics is reversible; the count is monotone; the only step that breaks the symmetry is the partial trace. The Hahn echo is the experimentum crucis: it separates the part of dephasing that leaves the spin–bath state a product (reversible, refocusable) from the part that entangles spins with the bath (irreversible, committed). This answers the objection "the monotonicity is true by definition, not earned from reversible mechanics."

Headline (Section 5): a perturbation-independent floor. Finally we count the distinguishable many-body trajectories a spin system can traverse in n coherent steps when it is coupled through d channels. The count is $T(n,d) = d(d+1)^{n-1}$, and its logarithmic growth rate per step, $\ln(d+1)$, depends only on the channel multiplicity d and *not* on the coupling strength. Converted to a physical rate through the local precession frequency, this predicts an irreducible Loschmidt-echo decay rate $1/\tau_{\mathrm{floor}} = (\omega_{\mathrm{loc}}/2\pi)\ln(d+1)$ that does not improve as the experimental reversal is made more perfect. This reproduces the central empirical finding of polarization-echo experiments (Jalabert and Pastawski, 2001; Levstein et al., 1998; Pastawski et al., 2000)—decay independent of the residual imperfection—and makes a prediction the semiclassical Lyapunov account does not: a nonzero floor even for integrable, non-chaotic spin systems.

1.3 Relation to existing theory

We are not proposing a replacement for relaxation theory; we are proposing a bound that the standard theory must respect and that, at saturation, the standard theory realizes. The relationship is the same as that between the Carnot bound and a specific heat engine: the bound is derived from a general principle (here, the quantum speed limit), and a particular microscopic model either saturates it or falls below it. The content of the paper is (a) that transverse relaxation *saturates* the counting ceiling in the rigid-lattice regime, (b) that the monotonicity of the count is forced by open-system structure rather than assumed, and (c) that one consequence—the echo floor—is both quantitative and, in a specific regime, distinguishable from the orthodox semiclassical prediction.

2 Setup and assumptions

We fix notation and state every assumption explicitly, so that each later theorem can be traced to the hypotheses it requires.

2.1 The spin–bath system

Definition 2.1 (Spin–bath decomposition)**.** The total Hilbert space is $\mathcal{H} = \mathcal{H}_S \otimes \mathcal{H}_B$, where $\mathcal{H}_S$ carries the observed spin ensemble (N spins, $\dim \mathcal{H}_S = 2^N$) and $\mathcal{H}_B$ carries everything else with which the spins interact (other spins, lattice phonons, molecular degrees of freedom). The total Hamiltonian is

$$H = H_S \otimes \mathbb{1} + \mathbb{1} \otimes H_B + H_{SB}, \tag{1}$$

with H_{SB} the spin–bath coupling. The reduced spin state is $\rho_S(t) = \mathrm{Tr}_B\, \rho(t)$, where $\rho(t)$ is the state of the closed total system.

Assumption 1 (Reversible substrate)**.** The total Hamiltonian (1) is time-independent and Hermitian, and is time-reversal invariant: there is an antiunitary Θ with $\Theta H \Theta^{-1} = H$. The total evolution $U(t) = e^{-iHt/\hbar}$ is unitary, hence exactly reversible.

This is the crux of Loschmidt's objection (Loschmidt, 1876): the substrate is reversible, and any irreversibility we derive must therefore not be smuggled in through a non-reversible postulate. Assumption 1 holds it fixed.

Assumption 2 (Initially uncorrelated preparation)**.** At $t = 0$ the spin ensemble is prepared in a definite state uncorrelated with the bath: $\rho(0) = \rho_S(0) \otimes \rho_B(0)$. In the spin-echo context this is the post-90°-pulse transverse state.

Assumption 2 is the standard and physically warranted initial condition for a relaxation experiment: the pulse prepares the spins independently of the bath microstate. It is the only place an "initial condition" enters, and it is the weakest possible one (a product state, not a low-entropy fine-tuning).

Assumption 3 (Coupling energy scale)**.** The spin–bath coupling has a finite energy spread in the relevant state,

$$\Delta E_{SB} := \big(\langle H_{SB}^2 \rangle - \langle H_{SB} \rangle^2\big)^{1/2} = \hbar\, \langle \delta\omega^2 \rangle^{1/2}, \tag{2}$$

where $\langle \delta\omega^2 \rangle$ is the second moment of the fluctuating local precession frequency. We write $b := \langle \delta\omega^2 \rangle^{1/2}$ for this coupling frequency; it is the single material parameter of the theory and is, in any concrete system, computable from the dipolar (or hyperfine, or exchange) couplings by the standard van Vleck moment formulae (Abragam, 1961).

2.2 The count

The central object is not the cycle count of a clock but a count of *distinguishable spin states the bath has come to encode.*

Definition 2.2 (Effective Schmidt rank / partition count)**.** Let $\rho_S(t)$ have von Neumann entropy $S_{vN}(\rho_S) = -\mathrm{Tr}\big(\rho_S \ln \rho_S\big)$. The *effective Schmidt rank* of the spin–bath state, equivalently the *partition count* of the spin ensemble, is

$$M_{\text{eff}}(t) := \exp\!\big[S_{vN}(\rho_S(t))\big]. \tag{3}$$

M_{eff} is the effective number of orthogonal bath states correlated with distinct spin states (the participation number of the Schmidt decomposition of the global pure state across the S|B cut) (Nielsen and Chuang, 2010).

Three remarks fix the interpretation. First, $M_{\text{eff}}(0) = 1$ by Assumption 2 (a product state has Schmidt rank one). Second, M_{eff} is dimensionless and basis-independent. Third, M_{eff} is exactly the quantity that controls echo attenuation: an echo recovers coherence only to the extent that the spin state can be disentangled from the bath by an operation on the spins alone, and the obstruction to that is precisely $S_{vN}(\rho_S) > 0$. We make this precise in Section 4.

Remark 2.3 (Why a count, and why this count). The word "count" is used advisedly. $\ln M_{\text{eff}} = S_{vN}(\rho_S)$ is the number of bits the bath has acquired about the spins; M_{eff} is the number of distinguishable alternatives those bits resolve. Relaxation, decoherence, and echo attenuation are all, in this reading, the growth of M_{eff}. The remainder of the paper bounds the *rate* of that growth, proves it *monotone*, and computes its *floor*.

3 Foundation: the counting ceiling on the relaxation rate

We first bound how fast M_{eff} can grow, using only the energy spread of the dynamics, and show that saturating the bound returns the known relaxation rate.

3.1 The Margolus–Levitin ceiling

Lemma 3.1 (Distinguishability rate ceiling)**.** *Let a quantum system evolve under a time-independent Hamiltonian of energy spread ΔE (with the ground-state energy set to zero, mean energy $\langle E \rangle$). The minimum time to evolve into an orthogonal (perfectly distinguishable) state satisfies the Mandelstam–Tamm and Margolus–Levitin bounds*

$$\tau_\perp \;\geq\; \max\left\{ \frac{\pi\hbar}{2\,\Delta E},\; \frac{\pi\hbar}{2\,\langle E \rangle} \right\}. \tag{4}$$

Consequently the rate at which mutually orthogonal states can be generated is bounded:

$$\frac{\mathrm{d}N_\perp}{\mathrm{d}t} \;\leq\; \frac{2\,\Delta E}{\pi\hbar}. \tag{5}$$

Proof. The Mandelstam–Tamm bound $\tau_\perp \geq \pi\hbar/2\Delta E$ follows from the energy–time uncertainty relation applied to the survival amplitude $\langle \psi(0)|\psi(t) \rangle$ (Deffner and Campbell, 2017; Mandelstam and Tamm, 1945); the

Margolus–Levitin bound $\tau_\perp \geq \pi\hbar/2\langle E\rangle$ is an independent inequality from the spectral decomposition of the survival amplitude (Margolus and Levitin, 1998). Both are tight. A sequence of $N_\perp$ mutually orthogonal states must be separated by at least $\tau_\perp$ in time, so in elapsed time t at most $t/\tau_\perp$ such states are reachable, giving (5). □

3.2 From the ceiling to the relaxation rate

We now apply Lemma 3.1 to the bath's acquisition of which-spin information. The relevant "system" for the speed limit is the *conditional bath state*: when the spins are in configuration s, the bath evolves under an effective Hamiltonian $H_{\mathrm{B}}^{(s)} = H_{\mathrm{B}} + \langle s|H_{\mathrm{SB}}|s\rangle$. Two distinct spin configurations $s \neq s'$ drive the bath along trajectories whose conditional Hamiltonians differ by $\Delta H_{\mathrm{B}} = \langle s|H_{\mathrm{SB}}|s\rangle - \langle s'|H_{\mathrm{SB}}|s'\rangle$, of energy spread of order ΔE_{SB} (Assumption 3).

Theorem 3.2 (Counting bound on transverse relaxation)**.** *Under Assumptions 1–3, the rate of growth of the partition count is bounded by the Margolus–Levitin ceiling at the coupling energy:*

$$\frac{\mathrm{d}}{\mathrm{d}t}\ln M_{\mathrm{eff}}(t) = \frac{\mathrm{d}S_{\mathrm{vN}}(\rho_{\mathrm{S}})}{\mathrm{d}t} \leq \frac{2\,\Delta E_{\mathrm{SB}}}{\pi\hbar} = \frac{2}{\pi}b. \quad (6)$$

The transverse relaxation rate, defined operationally as the initial decay rate of the transverse coherence $\mathcal{C}(t) = |\operatorname{Tr}(\rho_{\mathrm{S}}(t)\,\sigma_+)|$, is bounded by the same ceiling,

$$\frac{1}{T_2} \leq \frac{2}{\pi}b = \frac{2}{\pi}\langle\delta\omega^2\rangle^{1/2}. \quad (7)$$

Proof. The bath states correlated with distinct spin configurations become distinguishable no faster than the ceiling (5) allows, with $\Delta E = \Delta E_{\mathrm{SB}}$. Each newly distinguishable bath state adds at most one unit to the participation number of the Schmidt decomposition, i.e. increments M_{eff} by at most a factor controlled by $\mathrm{d}N_\perp/\mathrm{d}t$; taking logarithms, $\mathrm{d}\ln M_{\mathrm{eff}}/\mathrm{d}t \leq 2\Delta E_{\mathrm{SB}}/\pi\hbar$, which is (6). For (7): transverse coherence is the off-diagonal weight of ρ_{S}, and its decay is bounded below in time (equivalently the rate bounded above) by the rate at which the bath resolves the spin's phase, which is exactly the distinguishability rate just bounded. Substituting $\Delta E_{\mathrm{SB}} = \hbar b$ gives (7). □

3.3 Saturation reproduces the Anderson–Kubo rate

The ceiling (7) is an inequality. Its physical content is fixed by asking when it is saturated.

Proposition 3.3 (Rigid-lattice saturation)**.** *In the slow-bath (rigid-lattice) limit, where the bath correlation time τ_c is long compared with $1/b$, the ceiling (7) is saturated up to an $O(1)$ geometric factor, and the relaxation rate is*

$$\frac{1}{T_2} \simeq \langle\delta\omega^2\rangle^{1/2} = b, \quad (8)$$

the Anderson–Kubo rigid-lattice second-moment rate (Anderson and Weiss, 1953; Kubo, 1962), with no fitted proportionality constant.

Proof. When $\tau_c \gg 1/b$ the conditional bath Hamiltonians are effectively static over the coherence time, so the phase accumulated by the spin is $\varphi(t) \approx \delta\omega\, t$ with $\delta\omega$ drawn from a distribution of width b. The coherence is the Gaussian average $\mathcal{C}(t) = \langle e^{i\delta\omega t}\rangle = \exp(-\frac{1}{2}b^2t^2)$, whose $1/e$ time is $\sqrt{2}/b$. Identifying the decay rate gives $1/T_2 \simeq b$, which saturates (7) to within the factor $\pi/2$ (the difference between the Gaussian $1/e$ time and the orthogonality time of the Margolus–Levitin bound). The motional-narrowing limit $\tau_c \ll 1/b$ falls strictly below the ceiling, $1/T_2 \simeq b^2\tau_c \ll b$, as the standard theory also gives (Kubo, 1962): the system is then far from saturating the speed limit, consistent with the inequality. □

Remark 3.4 (What has and has not been derived). Proposition 3.3 reproduces the known rigid-lattice rate from a speed-limit ceiling, fixing the proportionality constant by saturation rather than by fitting. It does not, by itself, exceed Bloch–Redfield theory: in the rigid-lattice limit both give $1/T_2 \sim b$, and in the motional-narrowing limit both give $1/T_2 \sim b^2\tau_c$. The value added at this layer is conceptual: the rate is exhibited as the saturation of a universal bound on how fast a bath can acquire information, not as the output of a particular correlation function. The genuinely new prediction appears only at the third layer (Section 5), where the bound is applied to many-body trajectory counting and yields a coupling-independent floor. The bridge to that result is the mechanism of the next section, which is required to know that the count grows monotonically at all.

4 Mechanism: monotonicity from the partial trace

The ceiling of Section 3 bounds how fast M_{eff} can grow. It does not say that M_{eff} grows, nor that it grows only forward. Loschmidt's objection is precisely that a reversible substrate should not single out a direction. We now show that the count is monotone non-decreasing under the coarse-grained dynamics, that this monotonicity is a theorem about the partial trace rather than a postulate, and that the spin echo realizes the distinction between the reversible and irreversible parts of dephasing exactly.

4.1 The reduced dynamics is a CPTP map

Lemma 4.1 (Reduced evolution is completely positive and trace-preserving)**.** *Under Assumptions 1–2, the map $\Lambda_t : \rho_{\mathrm{S}}(0) \mapsto \rho_{\mathrm{S}}(t)$ defined by*

$$\Lambda_t(\rho_{\mathrm{S}}(0)) = \operatorname{Tr}_{\mathrm{B}}\Big[U(t)\,\big(\rho_{\mathrm{S}}(0)\otimes\rho_{\mathrm{B}}(0)\big)\,U(t)^\dagger\Big] \quad (9)$$

is completely positive and trace preserving (CPTP), and admits a Kraus representation $\Lambda_t(\cdot) = \sum_k K_k(t)\,(\cdot)\,K_k(t)^\dagger$ with $\sum_k K_k^\dagger K_k = \mathbb{1}$.

Proof. This is the Stinespring/Kraus construction (Breuer and Petruccione, 2002; Nielsen and Chuang, 2010). With an initial product state (Assumption 2), let $\rho_B(0) = \sum_\mu p_\mu |\mu\rangle\langle\mu|$ in its eigenbasis. Then $K_k(t) = \sqrt{p_\mu}\,\langle\nu|U(t)|\mu\rangle$ with composite index $k = (\mu, \nu)$ are operators on $\mathcal{H}_S$, unitarity of U gives $\sum_k K_k^\dagger K_k = \mathbb{1}$, and (13) is recovered. Complete positivity and trace preservation are immediate from the Kraus form. □

Remark 4.2 (Where the arrow enters). Lemma 4.1 is the entire source of irreversibility. The total map $\rho \mapsto U\rho U^\dagger$ is unitary and reversible (its inverse is $U^\dagger \cdot U$, equally physical). The reduced map Λ_t is in general *not* unitarily invertible: the partial trace Tr_B discards the spin–bath correlations, and once discarded they cannot be recovered by any operation on the spins alone. The arrow of time in this framework is neither postulated nor smuggled into the definition of a count; it is the non-invertibility of the partial trace, the one operation in the construction that fails to commute with time reversal.

4.2 Monotonicity of the count

Theorem 4.3 (Monotone partition count)**.** *Let the bath be initially in a maximally mixed (infinite-temperature) state, $\rho_B(0) \propto \mathbb{1}$, as is standard for nuclear spin baths at ordinary temperatures (Abragam, 1961). Then the partition count is non-decreasing along the coarse-grained evolution:*

$$M_{\text{eff}}(t_2) \;\geq\; M_{\text{eff}}(t_1) \qquad \text{for all } t_2 \geq t_1, \tag{10}$$

equivalently $S_{\text{vN}}(\rho_S(t_2)) \geq S_{\text{vN}}(\rho_S(t_1))$.

Proof. For a maximally mixed bath, Λ_t is a *unital* CPTP map (it preserves the maximally mixed system state, since the global dynamics cannot concentrate probability when the bath supplies no bias). Unital CPTP maps are entropy non-decreasing: by the monotonicity of relative entropy under CPTP maps (Lindblad, 1975) applied to the maximally mixed reference $\pi = \mathbb{1}/d_S$,

$$S(\Lambda_t\rho_S \,\|\, \pi) \leq S(\rho_S \,\|\, \pi), \tag{11}$$

and since $S(\sigma\|\pi) = \ln d_S - S_{\text{vN}}(\sigma)$ for unital Λ_t, this rearranges to $S_{\text{vN}}(\Lambda_t\rho_S) \geq S_{\text{vN}}(\rho_S)$. Composing maps over $[t_1, t_2]$ (the family Λ_t is a one-parameter CP-divisible evolution in the Markovian regime; for the non-Markovian case see Remark 4.5) gives (10). Exponentiating yields the statement for M_{eff}. □

Corollary 4.4 (Reversible substrate, irreversible count)**.** *The total von Neumann entropy $S_{\text{vN}}(\rho(t))$ is constant (unitary invariance), while the reduced entropy $S_{\text{vN}}(\rho_S(t))$ is non-decreasing. The conserved global entropy is redistributed into spin–bath correlations; the monotone reduced entropy is the count. No contradiction with Assumption 1 arises, because the two entropies refer to different states.*

Remark 4.5 (Non-Markovian baths and the softening of strict monotonicity). For a genuinely non-Markovian bath the family $\{\Lambda_t\}$ need not be CP-divisible, and $S_{\text{vN}}(\rho_S(t))$ can transiently *decrease*—this is information backflow, the microscopic basis of partial echo revivals. We therefore state the honest form of the result: $M_{\text{eff}}(t)$ is monotone non-decreasing on timescales long compared with the bath correlation time τ_c, and admits bounded transient dips on shorter scales whose integrated effect is recovered by refocusing. Strict monotonicity holds in the Markovian (coarse-grained) limit; controlled non-monotonicity on the scale τ_c is not a defect of the theory but its prediction for when echoes can partially revive. This ties the monotonicity of the count directly to the refocusability structure of the next subsection.

4.3 The spin echo separates the two parts exactly

We now make precise the claim that the Hahn echo recovers exactly the part of dephasing that did not increment the count.

Theorem 4.6 (Echo refocuses the product-state part only)**.** *Write the dephasing of a spin as the sum of a static contribution (from a fixed offset $\Delta\omega$ that is a property of the spin's location, not of the bath) and a fluctuating contribution (from H_{SB}). Then:*

(i) The static contribution keeps the spin–bath state a product: M_{eff} is unchanged, $S_{\text{vN}}(\rho_S) = 0$ throughout. A $180°$ pulse at τ maps the accumulated phase $\varphi \mapsto -\varphi$ and the ensemble refocuses exactly at 2τ. The dephasing is reversible because no count was incremented.

(ii) The fluctuating contribution entangles the spin with the bath: M_{eff} grows, $S_{\text{vN}}(\rho_S) > 0$. A $180°$ pulse acts on the spin alone and cannot disentangle it from the bath; the residual coherence at 2τ is reduced by $\exp[-\Delta S_{\text{vN}}]$, where ΔS_{vN} is the entropy committed in $[0, 2\tau]$. The dephasing is irreversible because the count was incremented.

Proof. (i) A static offset commutes with the partial trace: the spin phase is a c-number function of a fixed parameter, so $U(t)$ restricted to that contribution acts as a local unitary on $\mathcal{H}_S$, leaving the state a product and $S_{\text{vN}}(\rho_S) = 0$. A local unitary ($180°$ pulse) inverts the phase, and continued evolution under the same static offset cancels it at 2τ, independently of the offset value. (ii) The fluctuating coupling H_{SB} generates genuine spin–bath entanglement (the conditional bath states correlated with distinct spin phases become non-identical, hence non-orthogonal-to-orthogonal as in Theorem 3.2), so $S_{\text{vN}}(\rho_S) > 0$. By monogamy of entanglement and the locality of the pulse (Coffman et al., 2000), no operation on $\mathcal{H}_S$ alone can reduce $S_{\text{vN}}(\rho_S)$; the committed correlation is inaccessible to refocusing, and the echo amplitude carries the factor $e^{-\Delta S_{\text{vN}}}$. Identifying $\Delta S_{\text{vN}} = 2\tau/T_2$ recovers the standard echo attenuation $e^{-2\tau/T_2}$. □

Remark 4.7 (The qualitative becomes quantitative). Theorem 4.6 converts the familiar statement "T_2^* is refocusable and T_2 is not" into a statement about a computable quantity: refocusable $\Leftrightarrow S_{\rm vN}(\rho_{\rm S}) = 0$ (product state), non-refocusable $\Leftrightarrow S_{\rm vN}(\rho_{\rm S}) > 0$ (entangled state), and the echo deficit is exactly $e^{-\Delta S_{\rm vN}}$ with $\Delta S_{\rm vN}$ the entropy committed to the bath. The committed entropy is not a free parameter: its rate is bounded by Theorem 3.2 and, in the rigid-lattice limit, saturates to b by Proposition 3.3. The identification $\Delta S_{\rm vN} = 2\tau/T_2$ is therefore a derived relation, $1/T_2$ being the committed-entropy rate.

5 Headline: a perturbation-independent Loschmidt-echo floor

We now reach the result that distinguishes the counting framework from the correlation-function account. The Loschmidt (polarization) echo reverses the *many-body* interaction, $H \to -H + \Sigma$, where Σ collects the residual un-reversed couplings. The empirical discovery (Jalabert and Pastawski, 2001; Levstein et al., 1998; Pastawski et al., 2000) is that, in the strongly-coupled regime, the echo-fidelity decay rate saturates at a value *independent of* $\|\Sigma\|$: improving the reversal does not slow the decay. We derive this saturation, and its rate, from trajectory counting.

5.1 Counting many-body trajectories

Definition 5.1 (Channel multiplicity)**.** At each coherent step the spin system can commit which-state information through any of d independent coupling channels (e.g. the distinct neighbours of a spin in a dipolar network, or the independent decay modes available to the local cluster). d is a structural property of the coupling graph, not of the coupling strength.

Theorem 5.2 (Trajectory-count growth)**.** *The number of mutually distinguishable labelled commitment-trajectories the system can traverse in n coherent steps over d channels is*

$$T(n,d) = \sum_{k=1}^{n} \binom{n-1}{k-1} d^k = d\,(d+1)^{n-1}. \qquad (12)$$

Its logarithmic growth rate per step is

$$\lambda_{\rm count} := \lim_{n\to\infty} \frac{1}{n} \ln T(n,d) = \ln(d+1), \qquad (13)$$

independent of any coupling strength.

Proof. A commitment-trajectory of n steps is an ordered composition of n into k parts (the run-lengths between channel switches), each part labelled by one of d channels. Compositions of n into exactly k positive parts number $\binom{n-1}{k-1}$ (stars and bars); each part carries one of d labels, contributing d^k. Summing, $T(n,d) = \sum_{k=1}^{n} \binom{n-1}{k-1} d^k = d \sum_{j=0}^{n-1} \binom{n-1}{j} d^j = d(1+d)^{n-1}$ by the binomial theorem. Then $\frac{1}{n} \ln T = \frac{1}{n}[\ln d + (n-1)\ln(d+1)] \to \ln(d+1)$. The result depends only on the combinatorial multiplicity d, with no reference to $H_{\rm SB}$'s magnitude or to Σ. ☐

5.2 The floor rate

Theorem 5.3 (Perturbation-independent echo floor)**.** *Let $\omega_{\rm loc}$ be the local precession (commitment) frequency, so that one coherent step occupies time $2\pi/\omega_{\rm loc}$. In the strongly-coupled regime the Loschmidt-echo fidelity decays no slower than the rate at which distinguishable trajectories proliferate,*

$$\boxed{\frac{1}{\tau_{\rm floor}} = \frac{\omega_{\rm loc}}{2\pi} \ln(d+1)} \qquad (14)$$

and this rate is independent of the residual perturbation $\|\Sigma\|$ used to engineer the reversal.

Proof. The fidelity $F(t) = |\langle\psi_0|e^{iH't/\hbar}e^{-iHt/\hbar}|\psi_0\rangle|^2$ is the squared overlap of the forward-evolved state with the imperfectly back-evolved state. By Theorem 4.6(ii), the recoverable part of the overlap is limited by the committed entropy $S_{\rm vN}(\rho_{\rm S})$, which grows as the system explores distinguishable commitment-trajectories. After n steps the number of such trajectories is $T(n,d)$, so $S_{\rm vN} \geq \ln(\text{effective participation})$ grows at rate $\lambda_{\rm count} = \ln(d+1)$ per step (Theorem 5.2). Converting steps to time through $\omega_{\rm loc}/2\pi$ steps per unit time gives the decay rate (14). Independence of $\|\Sigma\|$ is inherited from (13): the proliferation rate is set by the channel multiplicity d alone, so reducing the residual coupling (improving the reversal) does not reduce it. The reversal can recover the configurational overlap but not the trajectory-counting entropy, exactly as the Hahn echo recovers T_2^* but not T_2. ☐

Corollary 5.4 (Numerical value for a three-dimensional network)**.** *For a spin embedded in a three-dimensional dipolar network with effective channel multiplicity $d = 3$ (three independent angular commitment directions), $\ln(d+1) = \ln 4 \approx 1.386$, and*

$$\frac{1}{\tau_{\rm floor}} \approx 0.221\,\omega_{\rm loc}. \qquad (15)$$

Identifying $\omega_{\rm loc}$ with the second moment of the local field, $\omega_{\rm loc} = \langle\delta\omega^2\rangle^{1/2} = b$, gives $1/\tau_{\rm floor} \approx 0.22\,b$: the floor is a fixed fraction of the rigid-lattice linewidth, set by the network dimensionality.

5.3 Why this is not the semiclassical Lyapunov result

Remark 5.5 (Point of departure from the Lyapunov account). The semiclassical theory of the Loschmidt echo in classically chaotic systems (Gorin et al., 2006; Jalabert and Pastawski, 2001) also predicts a perturbation-independent decay, identifying the saturated rate with a classical Lyapunov exponent $\lambda_{\rm Lyap}$ of the underlying dynamics. Equations (13) and (14) agree with that prediction in form—a coupling-independent exponential rate—but differ in origin and, crucially, in domain. The counting rate $\ln(d+1)$ is a property of the *coupling-graph multiplicity*, not of dynamical instability. It is therefore nonzero for systems whose classical limit is integrable or non-chaotic, where $\lambda_{\mathrm{Lyap}} = 0$ and the semiclassical theory predicts no floor. This is a sharp, falsifiable point of departure: a discrete, integrable, or few-body spin cluster with no positive Lyapunov exponent should, on the counting account, still exhibit a Loschmidt-echo floor at rate $(\omega_{\mathrm{loc}}/2\pi)\ln(d+1)$ fixed by its connectivity d, whereas the Lyapunov account predicts the floor recedes to zero as the reversal is perfected.

Remark 5.6 (Consistency with the chaotic regime). When the spin network *is* chaotic, the connectivity-set rate $\ln(d+1)$ and the Lyapunov rate λ_{Lyap} need not coincide; the observed floor is then the slower-saturating of the two mechanisms (the system decoheres at whichever rate first exhausts recoverable overlap). In dense dipolar solids the two are typically of the same order—$\ln 4 \approx 1.4$ versus Lyapunov exponents of order unity in units of the local coupling—which is why the existing data (Levstein et al., 1998; Pastawski et al., 2000) are consistent with either reading. The discriminating experiments are therefore in the *non-chaotic* regime (Section 6).

6 Experimental tests

The framework makes three quantitative claims, each independently testable.

6.1 Test 1: rigid-lattice rate as a saturated ceiling

Proposition 3.3 predicts $1/T_2 \simeq b$ in the rigid-lattice limit with no fitted constant. This is already confirmed by the standard second-moment analysis of solid-state lineshapes (Abragam, 1961; Anderson and Weiss, 1953); the contribution here is interpretive (the rate is the saturated Margolus–Levitin ceiling), so the test is one of *consistency*: in every system where $\tau_c \gg 1/b$, the measured $1/T_2$ should equal the van Vleck second moment b to within the geometric $O(1)$ factor of Proposition 3.3, and should fall strictly below b as τ_c decreases into motional narrowing.

6.2 Test 2: the floor's connectivity dependence

Corollary 5.4 predicts $1/\tau_{\mathrm{floor}} = (\omega_{\mathrm{loc}}/2\pi)\ln(d+1)$. Two consequences are directly measurable. First, the floor rate should scale as $\ln(d+1)$ as the coordination number of the spin network is varied (e.g. by isotopic dilution, which reduces the number of active dipolar partners d): a plot of $\tau_{\mathrm{floor}}^{-1}/\omega_{\mathrm{loc}}$ against $\ln(d+1)$ should be a straight line of unit slope through the origin. Second, the floor should be *independent* of the quality of the magic-echo or time-reversal pulse sequence once the reversal is good enough to enter the saturated regime—reproducing the observed environment-independence (Pastawski et al., 2000) but now with a predicted magnitude.

6.3 Test 3: a floor in a non-chaotic system (the decisive test)

Remark 5.5 gives the experiment that separates the counting account from the Lyapunov account. Prepare a small, well-characterized, *non-chaotic* or integrable spin cluster (for instance a linear chain with nearest-neighbour-only XY coupling, or a few-spin star geometry) whose classical limit has $\lambda_{\mathrm{Lyap}} = 0$. Perform a polarization-echo reversal of increasing quality.

- *Lyapunov prediction:* the echo decay rate $\to 0$ as $\|\Sigma\| \to 0$; the floor recedes.
- *Counting prediction:* the echo decay rate saturates at $(\omega_{\mathrm{loc}}/2\pi)\ln(d+1)$ with d the cluster's connectivity; the floor remains.

Observation of a nonzero, connectivity-set floor in a provably non-chaotic cluster would confirm the counting mechanism and exclude the pure Lyapunov account; observation of a receding floor would refute the counting mechanism at this level. Either outcome is informative.

7 Discussion

7.1 What resolves the Loschmidt objection, and what does not

The Loschmidt objection is that a time-reversal-invariant substrate cannot single out a temporal direction. The resolution offered here is precise about where the asymmetry lives. It is not in the substrate—Assumption 1 keeps the total Hamiltonian reversible and Corollary 4.4 keeps the total entropy constant. It is not in a definition—the count M_{eff} is the effective Schmidt rank of a genuine quantum state, and its monotonicity is a theorem (Theorem 4.3), not a stipulation. The asymmetry enters at exactly one place: the partial trace Tr_{B} (Remark 4.2), the operation that discards spin–bath correlations and that alone fails to commute with time reversal. The second law, in this reading, is the statement that the reduced count cannot decrease because the discarded correlations cannot be recalled by local operations.

This is a conventional resolution in its physics—it is the open-systems, decoherence-theoretic account (Breuer and Petruccione, 2002; Zurek, 2003)—but the paper makes two contributions beyond restating it. First, it fixes the *rate* of the count's growth by a universal speed-limit ceiling (Theorem 3.2) rather than by a model-specific correlation function, and shows the relaxation rate is its saturation (Proposition 3.3). Second, it derives a *coupling-independent echo floor* (Theorem 5.3) from pure trajectory counting, with a connectivity-set magnitude and a falsifiable prediction in the non-chaotic regime (Test 3).

7.2 Honest limitations

We list the load-bearing soft spots so they can be attacked directly.

1. *The step-to-time conversion.* Equation (14) identifies the coherent-step rate with $\omega_{\text{loc}}/2\pi$ and, in Corollary 5.4, ω_{loc} with the second-moment frequency b. This identification is physically motivated but not derived from first principles; a factor-of-order-unity geometric correction is possible and must be pinned down against data before the numerical coefficient 0.22 is treated as exact.

2. *The channel multiplicity d.* The value $d = 3$ (Corollary 5.4) is the natural choice for an isotropic three-dimensional dipolar network (three independent angular commitment directions), but the effective d for a given lattice is a modelling choice that the connectivity-scaling test (Test 2) is designed to calibrate, not assume.

3. *Strict versus coarse-grained monotonicity.* Theorem 4.3 gives strict monotonicity only in the Markovian limit; non-Markovian baths permit bounded transient decreases of the count (Remark 4.5). The framework predicts these as partial echo revivals rather than excluding them, but the claim "the count never decreases" must be qualified to "the count never decreases on scales long compared with τ_c."

4. *Saturation, not equality.* The relaxation rate equals the ceiling only at saturation (Proposition 3.3); away from the rigid-lattice limit the ceiling is a strict inequality and the actual rate is set by the bath spectral density, as in the standard theory. The counting framework bounds and, in one limit, reproduces the standard result; it does not replace it everywhere.

7.3 Outlook

The decisive next step is Test 3: a Loschmidt-echo measurement on a provably non-chaotic spin cluster. The counting framework predicts a nonzero floor fixed by connectivity where the semiclassical Lyapunov account predicts none. If the floor is observed with the predicted connectivity dependence, the relaxation rate, the echo deficit, and the many-body fidelity floor are unified as three manifestations of a single counting bound saturating a quantum speed limit. If it is not, the trajectory-counting layer (Section 5) is falsified while the ceiling (Section 3) and the monotonicity mechanism (Section 4) stand independently.

8 Conclusion

We have given an independent, information-theoretic account of transverse relaxation and echo attenuation built on three layers. A counting ceiling derived from the Margolus–Levitin quantum speed limit bounds the rate at which a bath can acquire which-spin information, and its saturation reproduces the Anderson–Kubo rigid-lattice relaxation rate $1/T_2 \simeq \langle \delta\omega^2 \rangle^{1/2}$ with no fitted constant. The monotonicity of the underlying count is derived from the non-invertibility of the partial trace—the arrow of time entering through open-system coarse-graining of a strictly reversible substrate, not through any postulate—and the Hahn echo is shown to separate the reversible (product-state) and irreversible (entanglement-committed) parts of dephasing exactly. Counting the distinguishable many-body commitment-trajectories then yields a perturbation-independent Loschmidt-echo floor $1/\tau_{\text{floor}} = (\omega_{\text{loc}}/2\pi)\ln(d+1)$, reproducing the observed environment-independence of polarization-echo decay and predicting—unlike the semiclassical Lyapunov account—a nonzero floor for non-chaotic spin systems. The framework is quantitative throughout, makes a falsifiable prediction that departs from the orthodox account in a specific regime, and locates the irreversibility of the spin echo in a single, identifiable place.

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